Prediction Chronic Kidney Disease From Symptoms

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Abstract

Chronic Kidney Disease (CKD) is a major global health concern that often progresses without noticeable symptoms until it reaches the advanced stages, which by then is too late. To prevent this, we focuses on early detection of CKD, which in turn relies on understanding how patient characteristics and clinical measurements contribute to disease risk. Using the medical dataset from the UCI Database containing laboratory values and symptom reports, this study examines the probabilistic relationships between CKD and its most influential predictors using the Expectation–Maximization (EM) algorithm.

1 Problem description

The general problem we are trying to analyze is how do risk factors (e.g. age, blood pressure, diabetes, hypertension) influence the likelihood of developing chronic kidney disease? Since Chronic Kidney Disease oftentimes develop silently within patients showing few symptoms until the disease has progressed to its late stages, it is important to recognize the little indicators that appears. Identifying individuals who are at high risk based on early clinical indicators, such as hypertension, diabetes, anemia, appetite loss, and age, is critical for timely intervention of severe outcomes like kidney failure. Understanding how these risk factors interact is essential because no single variable determines CKD on its own, instead, the condition arises from a complex interaction between many medical, demographic, and physiological features. To give a more focused path in our research, we will mainly examine: What is the probability of CKD given all of its direct parents (hypertension, diabetes, anemia, appetite, and age)?

$$P(\text{CKD} \mid \text{htn}, \text{dm}, \text{ane}, \text{appet}, \text{Age})$$

Then by the end of this, we will hope to be able to predict what is the predicted CKD probability (CKD = 1) for an old person (Age > 60) with diabetes (dm = 1), anemia (ane = 1), poor appetite (appet = 0), and hypertension (htn = 1). If given time, additional questions we wish to answer are:

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1. $P(\text{ane} = 1 \mid \text{htn} = 1, \text{dm} = 1, \text{age} = 20)$
2. $P(\text{htn} = 1 \mid \text{bgr} = 160)$
3. $P(\text{CKD} = 1 \mid \text{appet} = 0, \text{age} = 10)$

2 Data sourcing and processing

Our data is sourced from the UCI Machine Learning Repository. According to the dataset description, 400 instances of the data was collected from the hospital for a two-month period and was finally donated in 2015. The original dataset contains 24 features total and one label variable. To begin the preprocessing steps in our dataset, we selected only the top most relevent variables in our model to keep our belief network manageable in size and prevent issues like overfitting. Using advice from medical students and confirming it using medical sources, we narrowed it down to 13 variables:

- age (Age): this variables represented the patient's age in years, which matters in CKD risk because kidney function declines as people get older
- blood pressure (bp): this variable represented the patient's diastolic blood pressure measured in mm/Hg, which matters in kidney damage because high blood pressure stresses blood vessel
- sugar (su): this variable represented the amount of sugar present in the patient's body after fasting overnight, which can be an indicator of diabetes and a major risk for CKD
- red blood cells (rbc): this variable represented whether the red blood cells are normal or abnormal, which is important as abnormal red blood cells can be kidney damage
- blood glucose random (bgr): this variable represented the blood glucose measurement in mg/dl at random time of the day (no fasting is needed like the blood sugar test), which is important as high glucose level indicate a poor diabetic control that can harm kidney over time
- sodium(sod): this variable represent the sodium level in mEq/L, which is important because kidney regulate sodium balance, so abnormal level may suggest the kidney function has issue
- hypertension (htn): this variable represent whether or not the patient has hypertension, which matters as hypertension can cause or worsen kidney diseases
- diabetes mellitus (dm): this variable indicate whether a patient has diabete, which is important as diabete is the most common cause of chronic kidney diseases worldwide
- coronary artery disease (cad): this variable indicate whether a patient has heart disease, which can be a good indicator of kidney disease as poor cardiovascular health can stress kidney
- appetite (appet): this variable represent whether or not the patient has a normal or poor appetite as poor appetite is a symptom of worsening kidney function and metabolic imbalance
- pedal edema (pe): this function indicate if the patient is experiencing swelling in the feet, legs, or ankle caused by fluid retention, which can indicate kidney failure when excess fluid builds up in body
- anemia(ane): this variable indicates whether or not the patient is anemic, meaning their blood produces a less than normal amount of red blood cell. This can be an indicator of a damaged kidney as it will lead to a production of less erythropoietin that helps stimulating the bone marrow to produce more red blood cells.
- chronic kidney disease status (class): this variable indicates whether ot not the patient has chronic kidney disease, the target of our prediction model

Additionally, using medical knowlegde, we will perform discretization to turn continuous data such as age, blood pressure, blood glucose level, and sodium level into simple discrete values to keep things manageable for our analysis later. Based on the legal age classification, generally, you are considered young/young adult until you reach the age of 40, to which you are then classified as middle-aged until your 60s. As for the discretization of blood pressure, we know that there are two types of measurement, systolic, the higher number, and diastolic, the lower number. However, since

our dataset didn't specify which of the two types it is, we examined the values in the variable 'bp' and made a logical guess that it was the diastolic blood pressure, as most patients have relatively low readings. Using that knowledge, we researched the cutoff of high blood pressure for diastolic measurement and found that any number less than 80 and anything higher is high blood pressure (Stage 1 or Stage 2). Though there are more specific cutoffs that include 'elevated' category or 'hypertension crisis', we only stick to three categories as our dataset is relatively small and we don't want to overfit. For blood sugar (blood glucose random), we found from the CDC that any diastolic measure below 80 is good, but any greater than 90 is when it starts to get worse. As for sodium, we found that the healthy level is between 135 and 145. The following tables are a summary of how we split our data:

Table 1: Age Categories

Age
< 40 = young
40–60 = middle age
> 60 = senior

Table 2: Blood Pressure Categories

Blood Pressure
< 80 = normal
80–90 = stage 1
> 90 = stage 2

Table 3: Random Blood Glucose Categories

Blood Glucose (Random)
< 100 = normal
100–126 = prediabetic
> 126 = diabetic

Table 4: Sodium Levels

Sodium
< 135 = low
135–145 = normal
> 145 = high

Finally, since the dataset contains many missing values, we will not drop them, but will be using the EM algorithm as they are specifically designed to handle missing data and prevent bias of dropping missing values.

3 Modeling and inference

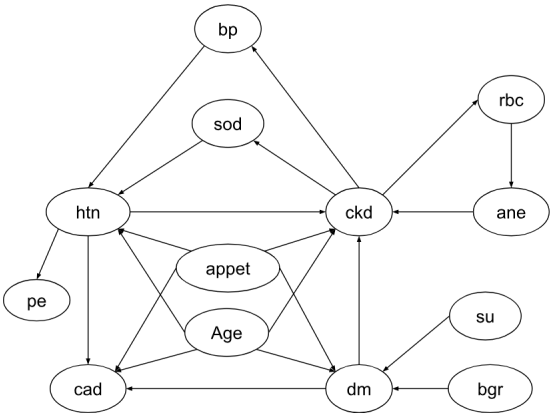


Figure 1: Predicted chronic kidney disease dependency chart.

We decided to model the relationships among patient attributes and Chronic Kidney Disease (CKD) using Bayesian Network (BN). As BN represents conditional dependencies between variables and provides a principled way to reason under uncertainty, this framework aligns well with our problem as CKD also depends on multiple interacting medical factors whose relationships are uncertain and potentially causal. Using purely research and medical knowledge, we created our initial belief network structure that we think will help move us forward in beginning to answer our research question, but due to several reasons we will explain later, it will be changed. For structure learning we decided to use a score-based strategy with Bayesian Information Criterion (mainly BIC-D for discrete values) because it helps prevent overfitting by comparing different statistical models and selecting the

best that balance goodness of fit and model complexity. We chose to use the Hill Climbing search algorithm alongside BIC because it is effective at identifying the best edges among a set of nodes, since it uses a greedy search strategy to iteratively select the best neighboring structure from the current one. After using hill climbing to get the basic directed acyclic graph (DAG) structure, we can learn the conditional probability tables (CPTs) using the expectation-maximization (EM) algorithm because we have missing values in the dataset and EM handles this data. For inference, we apply exact inference methods if the structure is small enough, however, if it becomes too complex we can use approximate inference. Some assumptions we make include the conditional independence relationships implied by the learned DAG is an approximation of the underlying relationships and variables are represented as discrete after preprocessing.

Referring back to Figure 1, we have a very loopy belief network, with several interdependencies (e.g. ckd $\rightarrow$ sod $\rightarrow$ htn $\rightarrow$ ckd), which is not allowed in a Bayesian Network structure. We had to either remove some edges or some variables before we can start running the hill climbing algorithm to improve our current model. While we did consider trimming down our dataset using d-separation and CKD's Markov blanket, in order to still use some of the non-loopy dependency in our original belief network, ultimately, we decided to rework the entire dependency completely, to keep all initial variables chosen (we believe they are all important). The reason why we did this is because of another benefit of using Hill Climbing. Even if we don't give it the initial edges to work with, it can simply find the dependency using just the given variables.

Before we started implementing our algorithms, we first created a 0.7 to 0.3 train-test split, so we have a test set to refer to see how well our algorithm performs. Then, we created functions to learn the structure of the graph using hill-climbing, fit our EM to the hill-climbed DAG using a Discrete Bayesian Network, and extract the conditional probability tables from the resulting EM.

Additionally, after parameter learning, missing values in the test set were imputed using probabilistic inference of observed values. For each observation, all non-missing features were treated as evidence, and variable elimination was used to compute the Maximum A Posteriori (MAP) assignment for any missing variables. These MAP estimates were then inserted back into the test matrix. This imputation procedure leverages the joint structure learned by the Bayesian Network during EM, ensuring that the inferred values are consistent with the model's conditional dependencies. This method also allowed us to predict the final ckd classification on the said test set simultaneously after we finish predicting all the missing values.

4 Results and Discussion

Table 5: Classification Performance Metrics with BIC-D

Metric	**Value / Description**
F1 score per class	[0.8842, 0.9241]
Micro-average F1	0.9083
	Computes F1 globally across all samples.
Macro-average F1	0.9042
	Averages F1 across classes without weighting.
Weighted-average F1	0.9102
	Weights class F1 scores by class frequency.
Precision	1.0

We trained the model using Hill Climbing with BIC-D scoring and EM, which provided a relatively stable results for our dataset of 400 instances. Before finally selecting a random state for our training-testing split to allow for reproduction of experiment, everytime we run the code, it returns a CPTs with some small changes to a few edges. However most edges, especially the strong ones (e.g. rbc $\rightarrow$ ckd, ckd $\rightarrow$ dm) stayed the same and the metric scoring only varies by a maximum ± 0.03 from our current result, which is still a good predictor. Although the dataset is relatively small and we believe that additional data would help improve the stability and the ability to learn dependencies and CPTs by providing more diverse variable combinations, having more data would also increase computational cost. Finding the DAGs with Hill Climbing, training EM, and then using variable elimination to impute data and predict CKD status will turn out to take hours in total rather than just

a few minutes. So with the current state of the structure learned with BIC-D on the training set, we applied the resulting classifier to the testing set and computed the F1 score for each class, along with the micro-average, macro-average, weighted-average F1 scores, and overall precision as shown above in table 5.

We chose to use F1 scoring to evaluate our model because of the imbalance between the two ckd classifications in our original dataset (250 positive, 150 negative). Purely evaluating our model based on its accuracy would have been misleading given the imbalanced dataset due to the fact that simply predicting positive every time would've resulted in a higher accuracy than purely random prediction (250/400 = 62.5%). Additionally, as we are working with medical data, there would likely be a significant cost to if we report false negatives. F1 scoring addresses these issues as it is meant to evaluate classification model performance on imbalanced datasets, using a harmonic mean of precision and recall in order to balance the importance of the two different metrics. The F1 score per class, $[0.8842, 0.9241]$, means that the model performs well on both the CKD and non-CKD classes, with relatively balanced precision–recall behavior. An F1 score of 0.8842 for non-CKD class indicates strong but not perfect accuracy in identifying non-CKD, while 0.9241 for CKD class shows even better consistency in correctly predicting CKD. Together, these values suggest that the classifier is able to distinguish CKD from non-CKD cases reliably, with only a small proportion of misclassifications for either group. The micro-average F1 score, 0.9083, means that when combining all predictions across both CKD and non-CKD classes, the model achieves strong overall performance, balancing precision and recall at the global level. It treats every individual prediction equally, so this value reflects how well the model performs overall, regardless of class imbalance. The macro-average F1 score, 0.9042, means that the model performs consistently well across both classes when each class is given equal importance. This score simply averages the F1 values of CKD and non-CKD, showing the model maintains good predictive ability even if one class is more frequent. The weighted-average F1 score, 0.9102, means that the model's performance is slightly boosted by the class with more samples. Since it weights each class's F1 score by its frequency, this score reflects the model's true performance on the dataset distribution, accounting for class imbalance. The precision score, 1.0, meant that every instance the model predicted as CKD was actually CKD, which meant the model made zero false positives. This is useful in medical settings where falsely labeling someone as having CKD can lead to unnecessary anxiety or follow-up tests.

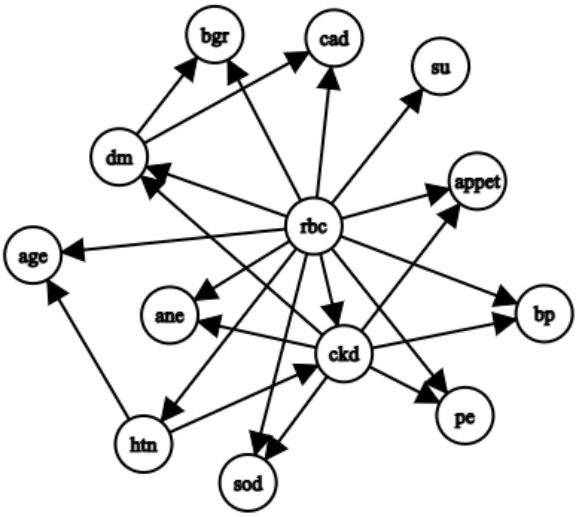


Figure 2: The learned DAG with BIC-D

Furthermore, the learned DAG in Figure 2 showed many important dependencies, including rbc being a parent to bgr, bp, sod, age, ckd, dm, htn, appet, su, pe, ane, and cad, therefore being a strong influence over multiple health variables. This aligns with medical intuition because low rbc is a consequence of CKD ("Anemia and Chronic Kidney Disease"), so the model captures the strength of the relationship rather than its causal direction. CKD is a parent of dm, pe, ane, appet, sod, and bp, implying that these variables may be consequences of CKD, which is reasonable because

CKD is known to result in pedal edema (Soi and Yee), anemia, appetite loss, and sodium imbalance (Zoccali et al.). High blood pressure can be both a cause as well as a consequence ("Chronic Kidney Disease and High Blood Pressure"). However, diabetes is often a cause of CKD rather than a result, which shows that the model misses the direction of causality with diabetes and rbc, but captures the correlations ("Diabetic Kidney Disease"). Moreover, htn is a parent of CKD, age, and cad, which is logical because hypertension is a critical risk factor for CKD ("High Blood Pressure & Kidney Disease") and CAD (Cubrilo-Turek).

While our current model with scoring method BIC-D is relatively good, we wanted to test it with other hyperparameters and see how well our current selection does with the others. Below is the performance of learning the CPTs based on other discrete scoring method:

Table 6: Classification Performance (K2)

Metric	Value
F1 score per class	[0.5638, 0.2857]
Micro-average F1	0.4583
Macro-average F1	0.4247
Weighted-average F1	0.3830
Precision	1.0

Table 7: Classification Performance (BDeu)

Metric	Value
F1 score per class	[0.8660, 0.9091]
Micro-average F1	0.8917
Macro-average F1	0.8875
Weighted-average F1	0.8940
Precision	1.0

Table 8: Classification Performance (BDs)

Metric	Value
F1 score per class	[0.8077, 0.8529]
Micro-average F1	0.8333
Macro-average F1	0.8303
Weighted-average F1	0.8371
Precision	1.0

Table 9: Classification Performance (AIC-D)

Metric	Value
F1 score per class	[0.8936, 0.9315]
Micro-average F1	0.9167
Macro-average F1	0.9126
Weighted-average F1	0.9182
Precision	1.0

Table 5 - 9 shows which model performs the best, which is AIC-D, and which performs the worst, K2. The reason why K2 performs the worst is because variable ordering is important in K2 to learn the correct graph. K2 strictly rely on predefined ordering of variables and it only consider parents set that comes before the node. However, since we simply pass in the original tables from Kaggle with some variables dropped and hadn't considered changing the column ordering, it perform terribly as a result because unlike the other methods, K2 doesn't consider reversing edges. AIC-D performs the best on the other hand because BIC-D is prone to producing very sparse result, which is good for generalizing, whereas AIC-D will seek for a model that minimizes information loss, sometimes overfitting model. Since our dataset is relatively small, because AIC-D is better than BIC-D, that means the BIC-D removes too many edges and that our dataset is not big enough yet to converge as explained earlier. Although AIC-D performs the best in the end because it is more flexible with small dataset, we still decided to choose BIC-D for the reason it yield a more interpretable network structure due to its sparse nature. Additionally, it imposes a stronger penalty on model complexity, helping us avoid overfitting and ensuring more stable EM parameter estimation. Since interpretability and robustness were our priorities, BIC-D was the more suitable choice for our analysis. Furthermore, BIC performs better overall on any size dataset, even the bigger ones. As the dataset size increase, the BIC will select the true model, but AIC, on the other hand, will overfit. So if we don't know if the dataset is relatively big or small, BIC-D is the most safest choice.

So using the model with BIC-D, we predicted the following probabilities for our research question:

1. $P(\text{CKD} = 1 \mid \text{htn} = 1, \text{dm} = 1, \text{ane} = 1, \text{appet} = 0, \text{Age} = \text{Senior}) = 1$
2. $P(\text{ane} = 1 \mid \text{htn} = 1, \text{dm} = 1, \text{age} = \text{Young}) = 0.2973$
3. $P(\text{htn} = 1 \mid \text{bgr} = \text{Diabetic}) = 0.3930$
4. $P(\text{CKD} = 1 \mid \text{appet} = 0, \text{age} = \text{Young}) = 0.6413$

First, the model predicts a 100% probability of CKD when a patient has hypertension, diabetes, anemia, poor appetite, and is a senior, indicating that this specific combination is an extremely strong indicator of CKD in this learned structure. In contrast, the model estimates that a young (20 year old) patient with hypertension and diabetes has a 29.73% chance of anemia, suggesting that while these

conditions increase risk, anemia is not guaranteed and is rather influenced by other factors. We also find that a patient with a diabetic blood glucose level (160) has a 39.30% chance of hypertension, reflecting the model's encoding of the relationship between elevated glucose and blood pressure. Finally, the model predicts a 64.13% probability of CKD for a young child (10 year old) with poor appetite, highlighting that appetite loss is a meaningful early signal even in younger patients, though the risk is lower than in older adults. Together, these results illustrate how the BIC-D model captures nuanced dependencies across clinical variables and allows probabilistic reasoning under different combinations of evidence.

5 Conclusion

Overall, the Bayesian Network model captured important probabilistic dependencies amongst the health variables and CKD. In particular, we can see red blood cell count being a strong predictor of CKD and also an influence of other health variables. Additionally, CKD is a parent to other complications like pedal edema, anemia, etc. The model achieved an F1 score of 0.917 for its weighted-average and perfect precision, thus, it is highly effective in predictive CKD from its risk factors and other variables. Some limitations include the size of the dataset, as it has only 400 instances, and this can misrepresent the causal direction for some variables, like diabetes. Additionally, the discretization of continuous variables may dilute certain relationships. Potential extensions or improvements include utilizing a larger dataset with more diversity and implementing alternative structure learning methods, such as constraint-based or hybrid methods, to capture the causal relationships that were missed.

6 Reflections & Contributions

One piece of advice we would give to future students working on a similar project is to take the time to thoroughly understand the dataset before building any models. Whether through background research or prior domain knowledge, knowing the relationships of each variable provides a strong foundation for meaningful analysis. This understanding prevents blindly relying on automated methods and it also helps us in being confident in the final model. In our project specifically, understanding the data, allowed us to confidently remove over ten unnecessary variables, reduce noise, and avoid overfitting. As a result, we achieved a high level of predictive accuracy while keeping the model interpretable and efficient. In this project we used Gen AI to help us with our research of the dataset by guiding us to the correct resources. The contribution of each member is as followed:

- Niroop: Helped with project brainstorming, help with researching different method, and help with coding. Through this project, I learned to apply Bayesian networks and the EM algorithm using pgmpy, gained experience with data processing in Python, and was introduced to new concepts such as search and score-based algorithms.
- Pratyush: Helped writing the skeleton code for applying bayesian networks at first which became the baseline for all future implementations and upgrades. I learned the proper way to do variable elimination and probabilistic imputation as a way to guess missing values.
- Nghi: Helped with tuning hyperparameters, help resolving code problems, and help with writeup. I learn how different scoring method works (K2, AIC-D, vs BIC-D) and more about how to learn a dag without an initial dag as we are given for homework.
- Nayana: Helped with initial brainstorming, data sourcing, figuring out the best models to use, research scoring method (BIC-D, BDeu) and searching algorithms (Hill Climbing), and contribute to write-up (results and discussion, conclusion). I learned how BNs, EM, and VE work in practice, how different score-based algorithms affect model performance, and where the model falls short in capturing directionality and relevant medical context.
- Vanessa: Helped with researching methods (e.g. implementation/application for imputation and f1) and contribute to LaTex Writeup. What I personally learned from the project was how to practically apply EM and use a Bayesian Network to learn from data and make predictions.

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